

Mini Project 2:
**Exploring Factors for Medical Student
Burnouts**

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1 Introduction

Burnout among medical students is a widespread and pressing concern, characterized by emotional exhaustion, cynicism, and a diminished sense of efficacy. Studies have estimated that up to half of medical students experience burnout during their training, a rate significantly higher than among their peers in other fields (Dyrbye et al. (2006)). Left unaddressed, burnout can lead to serious consequences including academic struggles, depression, anxiety, and an increased risk of dropping out. Given the demanding nature of medical education, the early identification of students at risk of burnout is critical for supporting their mental health.

Traditional tools such as the Maslach Burnout Inventory (MBI) are widely used to assess burnout symptoms in three dimensions: exhaustion, cynicism, and professional efficacy (Maslach et al. (1997)). However, these assessments are typically used reactively after burnout has already developed rather than as proactive tools for prevention. Leveraging machine learning techniques offers a promising approach for shifting from reactive assessment to proactive prediction by identifying students at risk before symptoms manifest.

This project aimed to build Random Forest models to predict burnout risk among medical students based on a comprehensive set of psychological, academic, and demographic features. Specifically, we focused on predicting the three sub-scales of burnout—emotional exhaustion, cynicism, and professional efficacy—as well as a combined total burnout score. We hypothesized that psychological distress variables, such as depression and anxiety, would emerge as the strongest predictors of burnout, and that different burnout dimensions would have distinct predictor profiles. We further hypothesized that feature selection would improve model performance by identifying a minimal set of high-impact predictors, demonstrating that a small subset of core features could achieve strong predictive accuracy.

2 Methods

2.1 Data

The dataset included survey responses from 886 medical students, with variables spanning psychological measures, academic factors, and demographics Devastator (2023). See the full list of variables in the Appendix - Section 6. Mental health outcomes are measured using validated scales, including the Maslach Burnout Inventory (MBI) for burnout (with subscales for exhaustion, cynicism, and professional efficacy), the CESD-R for depressive symptoms, and the STAI-T for anxiety. Additional scales capture empathy through the Questionnaire of Cognitive and Affective Empathy (QCAE) and job satisfaction. (Maslach et al. (1997)).

2.2 Exploratory Data Analysis

Before conducting the statistical analysis, an exploratory data analysis was performed to assess the initial relationships among variables. We first checked for missing values. There were no missing values in the data, as seen in Figure 1. This avoided the need for imputation or further data manipulation.

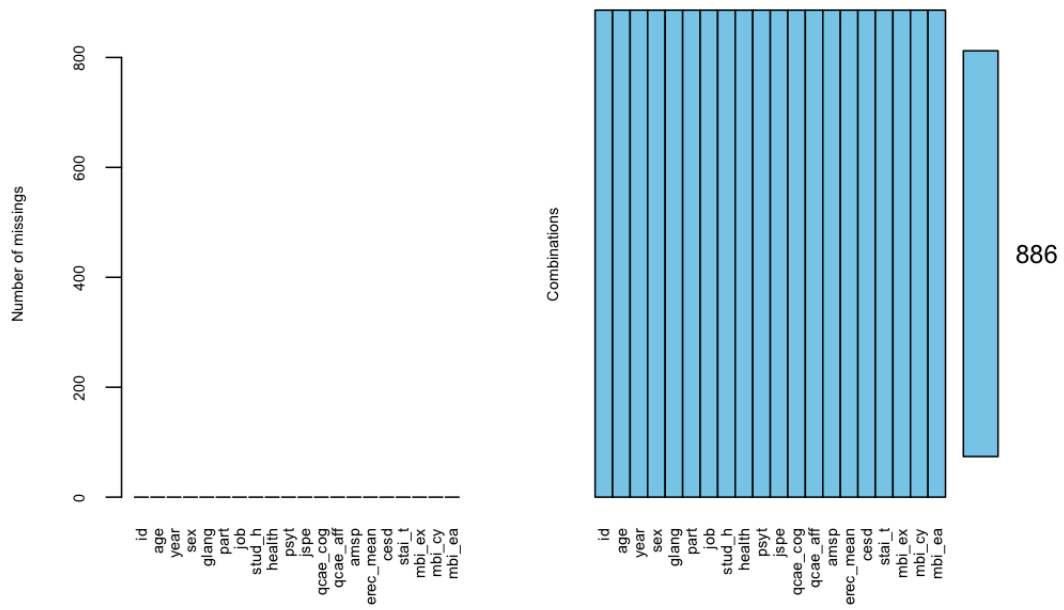


Figure 1: Plot of missing data.

To explore the relationship among the different features, we ran a general correlation matrix (as shown in Figure 2) that depicts the bivariate associations across all the features of the dataset. This revealed how depression and anxiety are positively related to exhaustion and cynicism and negatively related to efficiency.

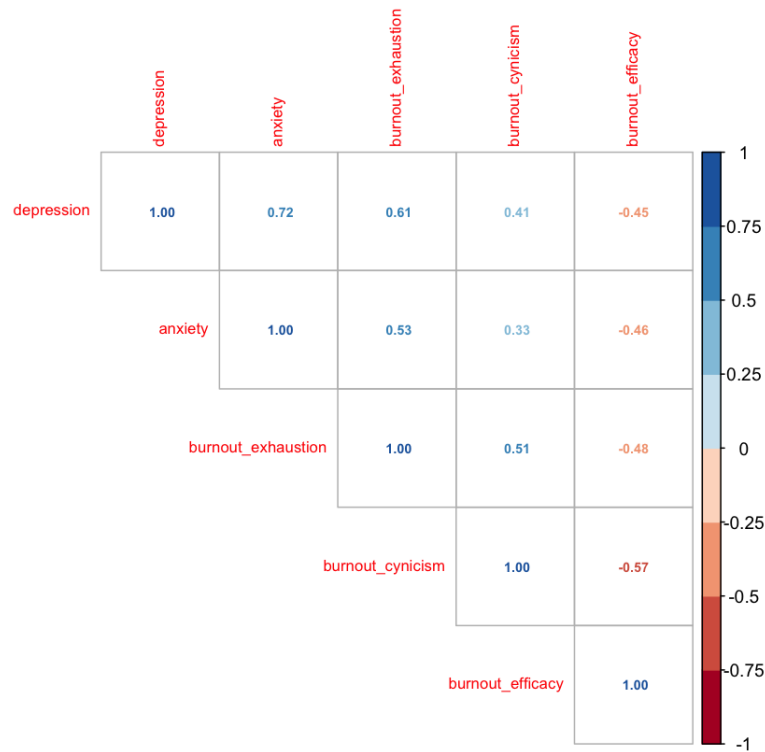


Figure 2: A correlation matrix of the different features in the dataset.

Because of this preliminary analysis, we deepened our focus on the specific relationships between each dimension of burnout and psychological distress, namely, depression and anxiety, divided by biological sex.

As depicted in Figures 3, 4, 5 and 6, in all cases, both male and female students demonstrated significant positive correlations between anxiety, depression, and elevated burnout levels. The efficacy plot showed a positive association with psychological distress after rescaling. For both male and female students, increased depressive or anxious symptoms corresponded to diminished perceptions of professional efficacy and heightened exhaustion and cynicism.



Figure 3: Impact of Anxiety and Depression on Efficacy by Sex.

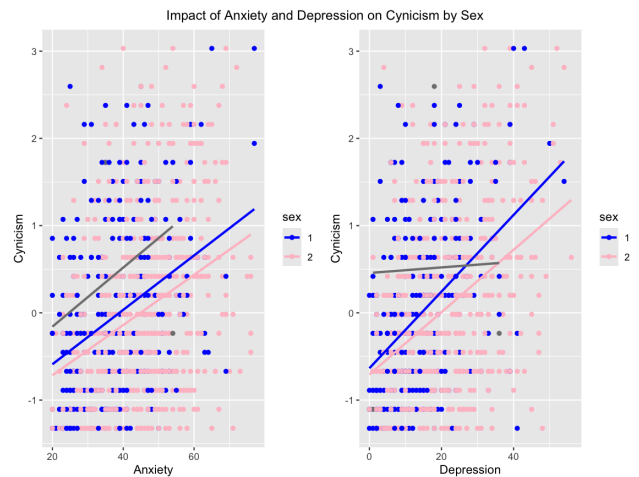


Figure 4: Impact of Anxiety and Depression on Cynicism by Sex.

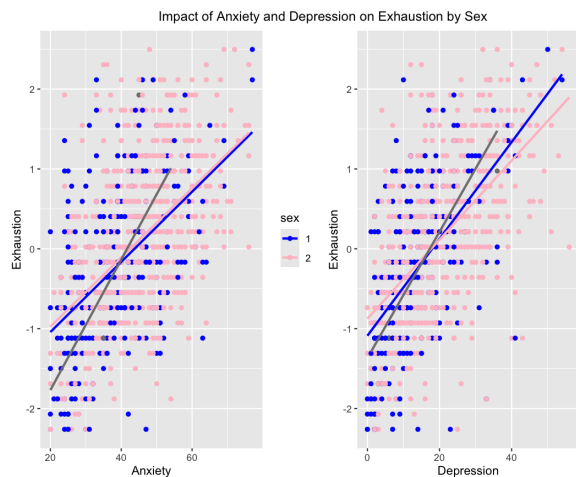


Figure 5: Impact of Anxiety and Depression on Exhaustion by Sex.

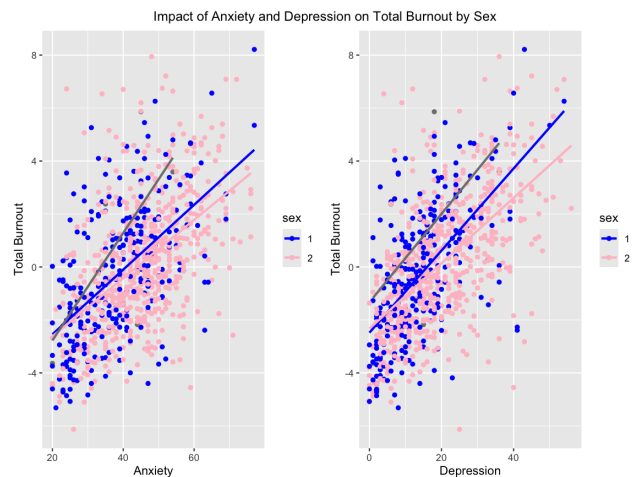


Figure 6: Impact of Anxiety and Depression on Total Burnout by Sex.

Moreover, we explored how the level of burnout related to the years of studying. Interestingly, the results were different for each type of burnout. The graphs in Figures 7, 8, 9 and 10 illustrate that professional efficacy remained relatively stable throughout the academic years with the exception of a decrease in efficacy during the second year, while cynicism exhibited a marked increase beginning in the third year of study. In contrast, symptoms of exhaustion tended to decrease as students advanced. This preliminary analysis showed us that overall, the levels of total burnout remained constant dur-

ing the years of study with a slight decrease in the last year, but the subsets of burnout (exhaustion, cynicism, and efficacy) demonstrated divergent trajectories.

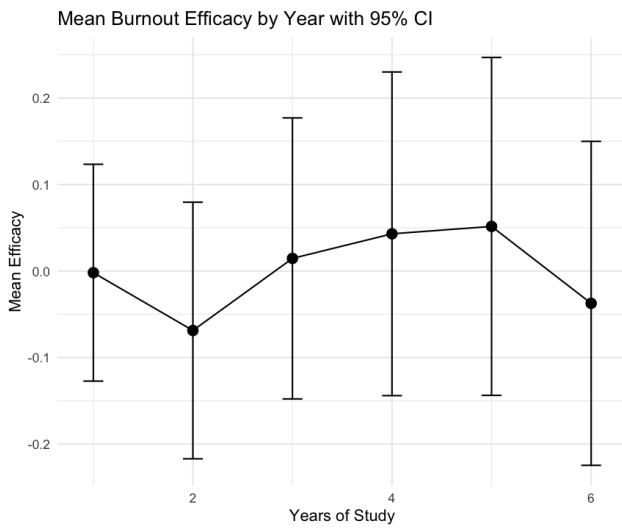


Figure 7: Mean Burnout Efficacy by Year with 95% CI.

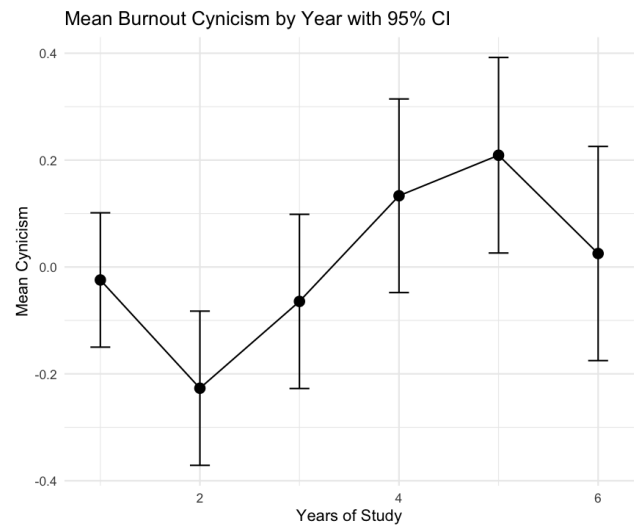


Figure 8: Mean Burnout Cynicism by Year with 95% CI.

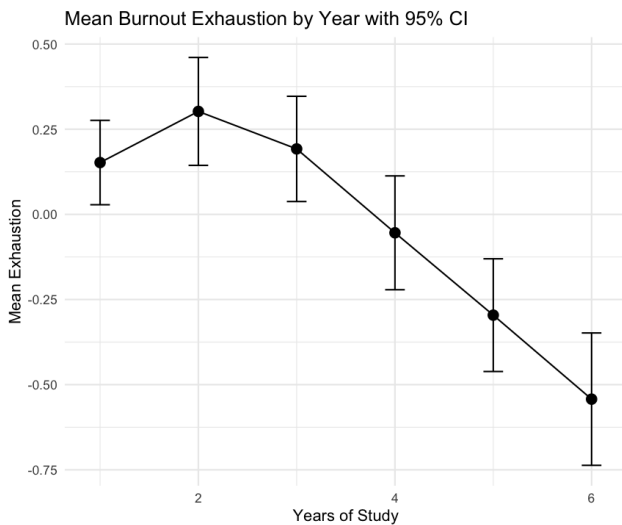


Figure 9: Mean Burnout Exhaustion by Year with 95% CI.

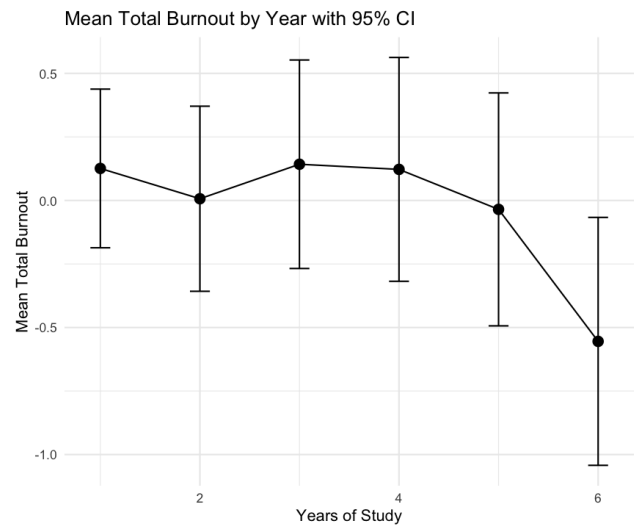


Figure 10: Mean Total Burnout by Year with 95% CI.

2.3 Cleaning and Preprocessing

To begin the data cleaning and preprocessing steps we removed the 'ID' information and renamed the columns for clarity. Then, we prepared the burnout variables for analysis. The dataset contained three individual measures of burnout: exhaustion (mbi_ex), cynicism ('mbi_cy'), and personal efficacy ('mbi_ea'). We reverse-coded the professional efficacy scale so that higher values uniformly indicated greater burnout, as with the other measures. This was done by subtracting each respondent's efficacy score from the maximum observed value in the dataset. Next, we standardized each of the three burnout dimensions using z-score transformation. Z-scoring transforms variables to have a mean of zero and a standard deviation of one. This procedure was applied to the burnout scores to place them on a

common scale, ensuring that each contributed equally.

With each individual measure scaled, we created one feature subset with the common variables and individual dataframes for each burnout scale. The feature subset included all relevant demographic, occupational, psychological, and affective variables associated with burnout. We also created a new measure called 'total_burnout' which summed the standardized exhaustion, cynicism, and personal efficacy values into a new combined measure of burnout. A dataframe was also created for the total burnout measure. We are left with four new dataframes (df_exhaustion, df_cynicism, df_efficacy, df_all_burnout) to use in our models.

2.4 Random Forest and Feature Selection

Next we trained four random forest models to predict burnout, one for each subscale (exhaustion, cynicism, and professional efficacy) and for the composite total burnout score. Each model was fit with all predictors. The models were implemented using the randomForest package in R, with 500 trees and default hyperparameters. We included the importance = TRUE option to compute feature importance based on the mean decrease in mean squared error (MSE). This allowed us to identify which variables most strongly influenced each burnout outcome. A fixed random seed was used to ensure reproducibility. Separate models were trained for each outcome to allow for direct comparison of predictor relevance across burnout dimensions, as well as to identify whether distinct psychological or contextual factors predicted specific types of burnout.

We evaluated the importance of each predictor in the random forest models using the percent increase in mean squared error (%IncMSE). This measures how much of the model's prediction error increases when a variable is permuted. Our goal with this was to identify which predictors were most influential to predicting burnout. We assumed that each subsection would have different influential predictors. Identifying this could inform future targeted interventions that minimize burnout among medical students.

Figure 11 shows the plot of influential variables for the exhaustion model. Depression and anxiety were the most influential variables, followed by academic year, age, and self-rated health. Figure 12 shows that for the cynicism model, depression and anxiety again ranked highest, reinforcing their consistent role across burnout dimensions. Additionally, study hours, academic year, and age emerged as important contributors. For the efficacy model, shown in figure 13 depression, anxiety, and study hours emerged again as the top predictors. Cognitive empathy, academic year, and sex were moderately important, while variables like job satisfaction, part-time status, and language background were less impactful. Finally, the total burnout model, which combined the three standardized subscales, reflected an aggregate importance profile. Once again, depression and anxiety dominated, followed by study hours, academic year, and cognitive empathy. Most of the remaining variables contributed little, indicating that a core set of psychological and academic workload factors drove total burnout predictions across models. A full list of the variables and their importance for each burnout model can be seen in Figure 20 in the Appendix - Section 6.

Collectively, these results suggest that emotional distress (depression and anxiety) and academic factors (year in program, study hours) were the most consistent and powerful predictors of burnout. In contrast, demographic features and traits related to empathy played a relatively minor role in predicting burnout.

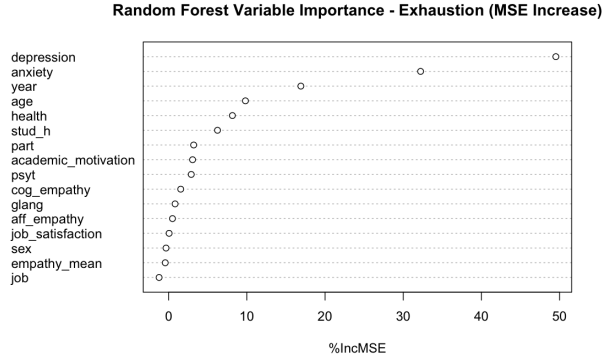


Figure 11: Variable importance from exhaustion model.

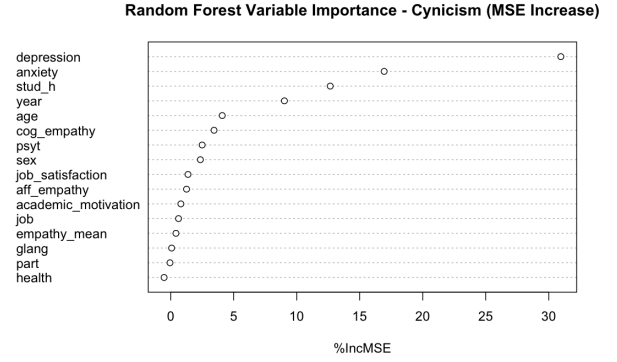


Figure 12: Variable importance from cynicism model.

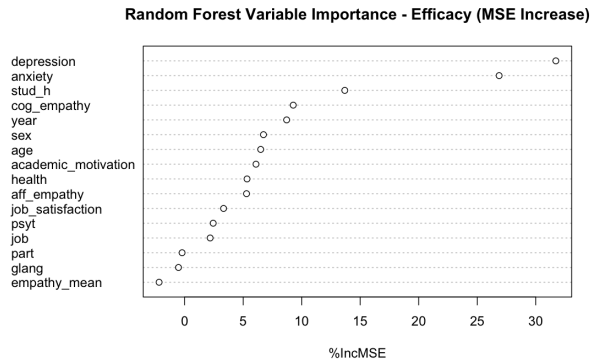


Figure 13: Variable importance from efficacy model.

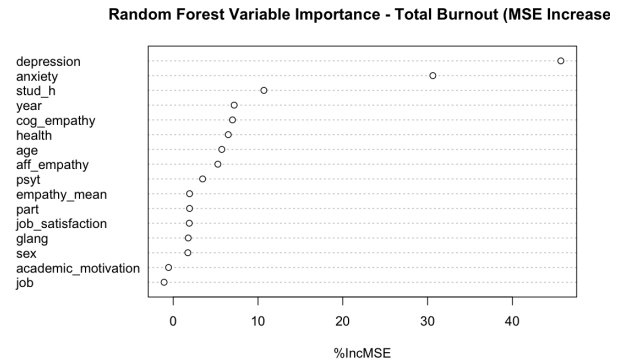


Figure 14: Variable importance from total burnout model.

2.5 Optimal Number of Features

To evaluate the predictive efficiency of individual variables and determine the optimal subset of features for each burnout outcome, we implemented a variable selection pipeline using Random Forests. Separate models were constructed for each target variable: exhaustion, cynicism, professional efficacy, and total burnout. For each model, we first computed variable importance using the %IncMSE metric. Predictor variables were then ranked in descending order of importance.

Using this ranked list, we implemented a forward selection loop, fitting the models with the top k variables (where $k = 1, 2, \dots, p$) in each iteration. At each step, the model was trained on 70% of the data and evaluated on the remaining 30% using mean squared prediction error (MSPE) as the performance metric. The resulting MSPE values were recorded and plotted against the number of included variables. The value of k that minimized out-of-sample MSPE was selected as the optimal model configuration. This procedure enabled data-driven feature selection and allowed us to assess the marginal contribution of each predictor to model performance while mitigating the risk of overfitting. We did this so that we could see not just the most important features but the optimal number of features for each predictive model. This approach also helped isolate which specific predictors were most closely associated with each burnout dimension, facilitating more interpretable and actionable models. Identifying the minimal set of high-impact features provides a foundation for developing targeted interventions tailored to the distinct drivers of exhaustion, cynicism, or diminished efficacy.

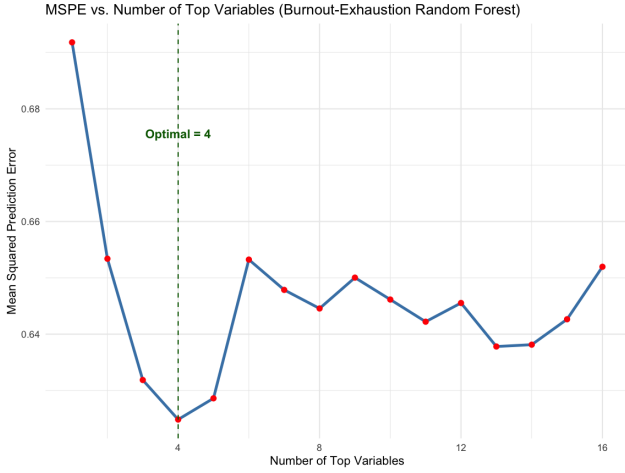


Figure 15: MSPE vs. Number of Top Variables (Burnout-Exhaustion Random Forest)

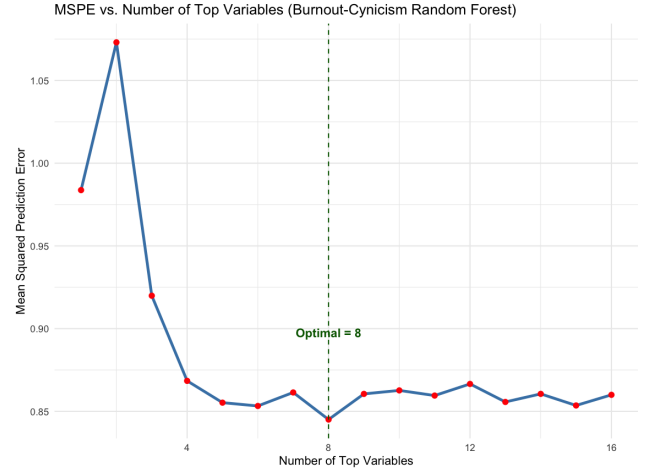


Figure 16: MSPE vs. Number of Top Variables (Burnout-Cynicism Random Forest)

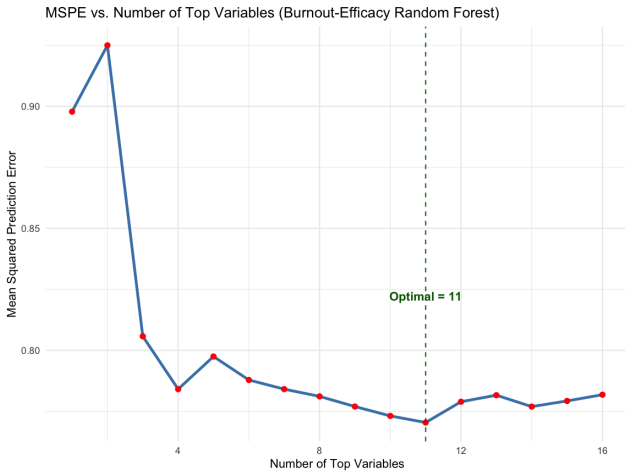


Figure 17: MSPE vs. Number of Top Variables (Burnout-Efficacy Random Forest)

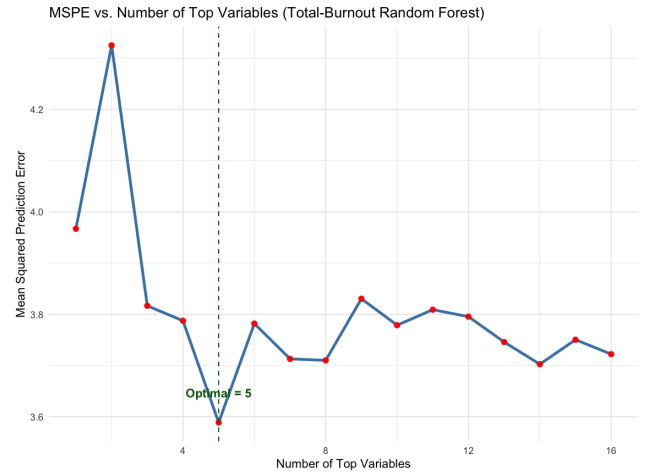


Figure 18: MSPE vs. Number of Top Variables (Total-Burnout Random Forest)

For the exhaustion model, Figure 15 shows that the MSPE decreases sharply with an increase in the top k variables from 1 to 4, and reaches its global minimum at $k = 4$. Beyond this point, the MSPE fluctuates slightly but does not improve substantially, indicating diminishing returns from including additional predictors. The dashed vertical line marks this optimal number of features ($k = 4$), which balances model simplicity and predictive accuracy. This also suggests that this model can perform just as well, if not better, using only the top 4 features rather than including all 16 features.

Figure 16 shows a steep descent in MSPE from 2 to 8 variables for the cynicism model, with the lowest MSPE achieved when using the top 8 predictors. This optimal point is highlighted by the vertical dashed line labeled "Optimal = 8". After this point, the MSPE remains relatively stable with minor fluctuations, suggesting that adding more features beyond 8 does not significantly enhance predictive performance.

Figure 17 for the efficacy model exhibits a sharp drop in MSPE when moving from 2 to 4 variables, after which there are no significant fluctuations. The lowest MSPE is achieved when using the top 11 variables, marked by the vertical dashed line labeled "Optimal = 11". Beyond this point, the performance does not improve noticeably. This result indicates that a relatively larger number of predictors

(compared to other models) is beneficial for modeling efficacy. The model appears to require more variables to minimize error, possibly due to the multidimensional nature of the efficacy construct.

In Figure 18 for total burnout, we see that there is an initial spike at $k = 2$, before the MSPE decreases. The lowest MSPE occurs when using the top 5 variables, indicated by the vertical dashed line labeled “Optimal = 5”. After this point, additional predictors do not meaningfully reduce prediction error, and the MSPE stabilizes with minor noise. This result suggests that while total burnout appears more difficult to predict (as seen in the higher overall MSPE values), a small subset of key variables can still provide strong predictive power. Including more features beyond the optimal set introduces unnecessary complexity without improving accuracy.

3 Results

3.1 Model Comparison

To evaluate the effectiveness of feature selection in improving model performance, we compared two versions of Random Forest models for each burnout dimension: one using the full set of predictors and one using an optimal subset of top-ranked variables identified by variable importance ranking.

Table 1 below summarizes the Mean Squared Prediction Error (MSPE) for both of these model types across the four burnout outcome variables: Exhaustion, Cynicism, Efficacy, and Total Burnout. Additionally, for each optimal model, the number of top predictors (K) that minimized the MSPE is reported.

Model	Version	MSPE	K
Exhaustion	Full	0.6519626	16
Exhaustion	Optimal	0.6248454	4
Cynicism	Full	0.8599912	16
Cynicism	Optimal	0.8450732	8
Efficacy	Full	0.7818373	16
Efficacy	Optimal	0.7704412	11
Total Burnout	Full	3.7222160	16
Total Burnout	Optimal	3.5887194	5

Table 1: The Mean Square Prediction Error (MSPE) for two different versions of each Random Forest for each burnout dimension.

Across all four burnout models, the optimal subset models consistently performed better than the full models, achieving lower MSPE values with fewer predictors:

- For the Exhaustion model, the subset with the top 4 predictors performed the best and lowered the MSPE from 0.652 to 0.635.
- The optimal 8-variable Cynicism model was slightly better in performance (MSPE = 0.846) than the full version (MSPE = 0.860).
- The smallest difference in performance was shown by the Efficacy submodel (MSPE = 0.770) with 11 most important predictors compared to the full version (MSPE = 0.782).

- The largest difference in performance was shown by the Total Burnout model where, when the top 5 predictors were used, the MSPE fell from 3.722 to 3.589.

These results indicate that model simplification via variable selection can enhance generalization performance of these random forest models while also reducing model complexity. None of the optimal models performed worse than their full counterparts, further reinforcing the value of targeted feature selection in predictive modeling.

3.2 Visualization

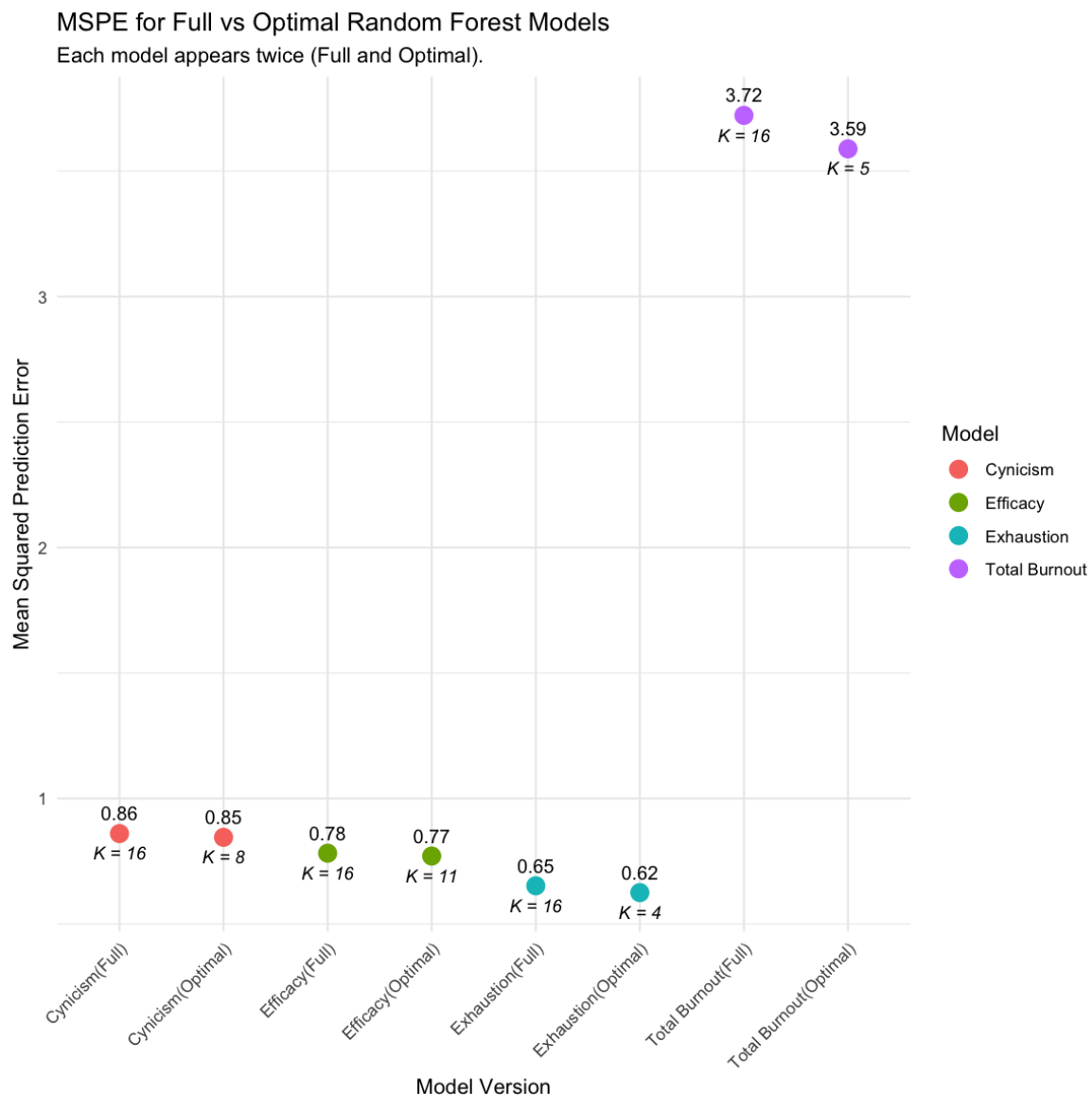


Figure 19: A comparison of the Mean Squared Prediction Error (MSPE) for the full and optimal-feature version of each Random Forest model across four burnout outcomes. Each outcome appears twice on the x-axis, once for the full model (using all 16 predictors) and once for the optimized model (using a subset of the top-ranked features).

The plot above visually represents the data in Table 1. For all 4 models the optimal-feature versions either outperformed or matched the full models in terms of MSPE. Each dot is labeled with its MSPE,

and the corresponding number of features (K) is annotated below. This visual comparison confirms that the selection of the targeted features not only simplifies the model but also maintains or enhances performance, making it a valuable approach in the predictive modeling of the dimensions of burnout.

4 Discussion

This study aimed to predict burnout among medical students using Random Forest models trained on psychological, academic, and demographic variables. Our results supported the initial hypotheses: psychological distress variables, particularly depression and anxiety, consistently emerged as the strongest predictors across all burnout models. This aligns with prior research highlighting the significant role of mental health challenges in shaping burnout risk among medical students (Dyrbye et al. (2006)).

Additionally, we observed that the distinct dimensions of burnout—exhaustion, cynicism, and professional efficacy—exhibited different predictor profiles. For example, the exhaustion model achieved optimal performance with just four predictors, while the efficacy model required a broader set of eleven variables to reach minimal error. This suggests that while psychological distress is a universal driver of burnout, certain facets, such as professional efficacy, may be influenced by a more complex combination of factors, including empathy, sex, and academic year. Feature selection proved to be an effective strategy for improving model performance. In all cases, the models trained on an optimal subset of predictors outperformed or matched the full models while reducing complexity. This demonstrates that a small set of key predictors can provide strong predictive accuracy, offering a more interpretable and actionable framework for identifying students at risk.

Our findings have several practical implications for medical education. The consistent influence of depression and anxiety underscores the importance of integrating mental health screenings into wellness programs. Academic factors, such as study hours and year in the program, also emerged as relevant predictors, suggesting that interventions should address both psychological and structural contributors to burnout. The ability to predict burnout risk using a small, targeted set of features could enable more efficient allocation of support resources, focusing interventions on the most vulnerable students.

However, this study also has limitations. The cross-sectional nature of the data prevents us from inferring causality, and the models were developed on a single dataset, which may limit generalizability. Future research could extend this work by applying the models to other medical student populations or incorporating longitudinal data to track how burnout risk evolves over time. Additionally, while Random Forests provide strong predictive performance, exploring alternative models such as gradient boosting or neural networks could yield further insights.

Overall, our analysis demonstrates the feasibility and value of using machine learning approaches for early identification of burnout risk in medical students. By combining data-driven feature selection with interpretable models, this approach sets up the development of targeted interventions that promote student well-being and success throughout their medical education.

5 Bibliography

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- Liselotte N. Dyrbye, Matthew R. Thomas, Jeffrey L. Huntington, Karen L. Lawson, Paul J. Novotny, Jeff A. Sloan, and Tait D. Shanafelt. Personal Life Events and Medical Student Burnout: A Multicenter Study. *Academic Medicine*, 81(4):374, April 2006. ISSN 1040-2446. URL https://journals.lww.com/academicmedicine/abstract/2006/04000/personal_life_events_and_medical_student_burnout_.10.aspx.
- Christina Maslach, Susan Jackson, and Michael Leiter. The Maslach Burnout Inventory Manual. In *Evaluating Stress: A Book of Resources*, volume 3, pages 191–218. The Scarecrow Press, January 1997. Journal Abbreviation: *Evaluating Stress: A Book of Resources*.

6 Appendix

Columns Present in the Medical Students Burnout Dataset

- id: Participants ID number.
- age: Age of the participant. (Integer)
- year: Year of study of the participant. (Integer)
- sex: Gender of the participant. (String)
- glang: Language spoken by the participant. (String)
- part: Whether the participant has a partner or not. (Integer)
- job: Job of the participant. (String)
- stud_h: Hours of study per week of the participant. (Integer)
- health: Self-reported health status of the participant. (String)
- psyt: Psychological distress score of the participant. (Integer)
- jspe: Job satisfaction score of the participant. (Integer)
- qcae_cog: Cognitive empathy score of the participant. (Integer)
- qcae_aff: Affective empathy score of the participant. (Integer)
- amsp: Academic motivation score of the participant. (Integer)
- erc_mean: Empathy rating score mean of the participant. (Integer)
- cesd: Center for Epidemiologic Studies Depression scale of the participant. (Integer)
- stai_t: State-Trait Anxiety Inventory scale of the participant. (Integer)
- mbi_ex: Maslach Burnout Inventory-Exhaustion scale of the participant. (Integer)
- mbi_cy: Maslach Burnout Inventory - Cynicism Scale of the participant. (Integer)
- mbi_ea: Maslach Burnout Inventory - Professional Efficacy Scale of the participant. (Integer)

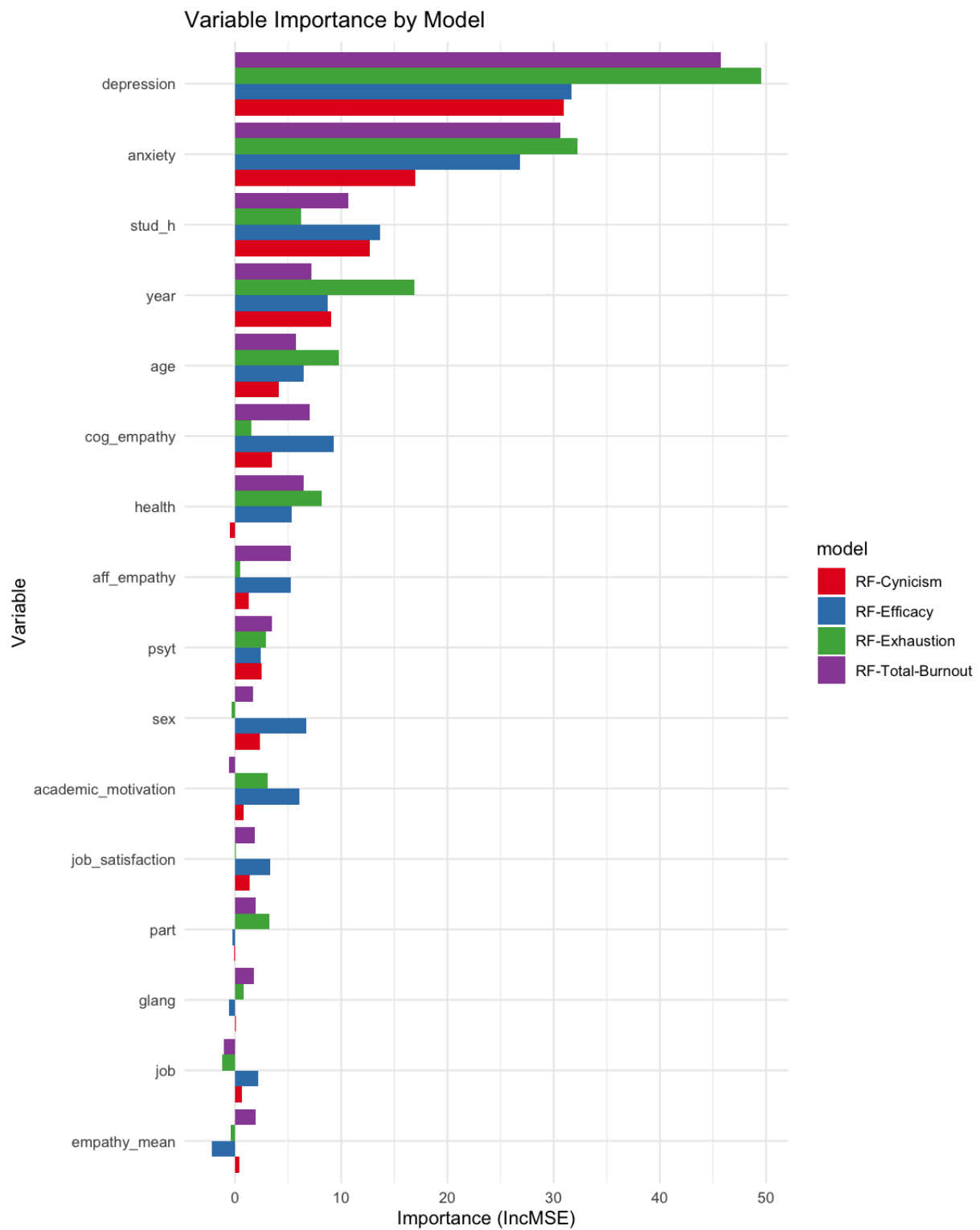


Figure 20: Variable Importance by each Burnout Model.